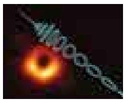
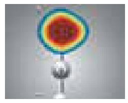


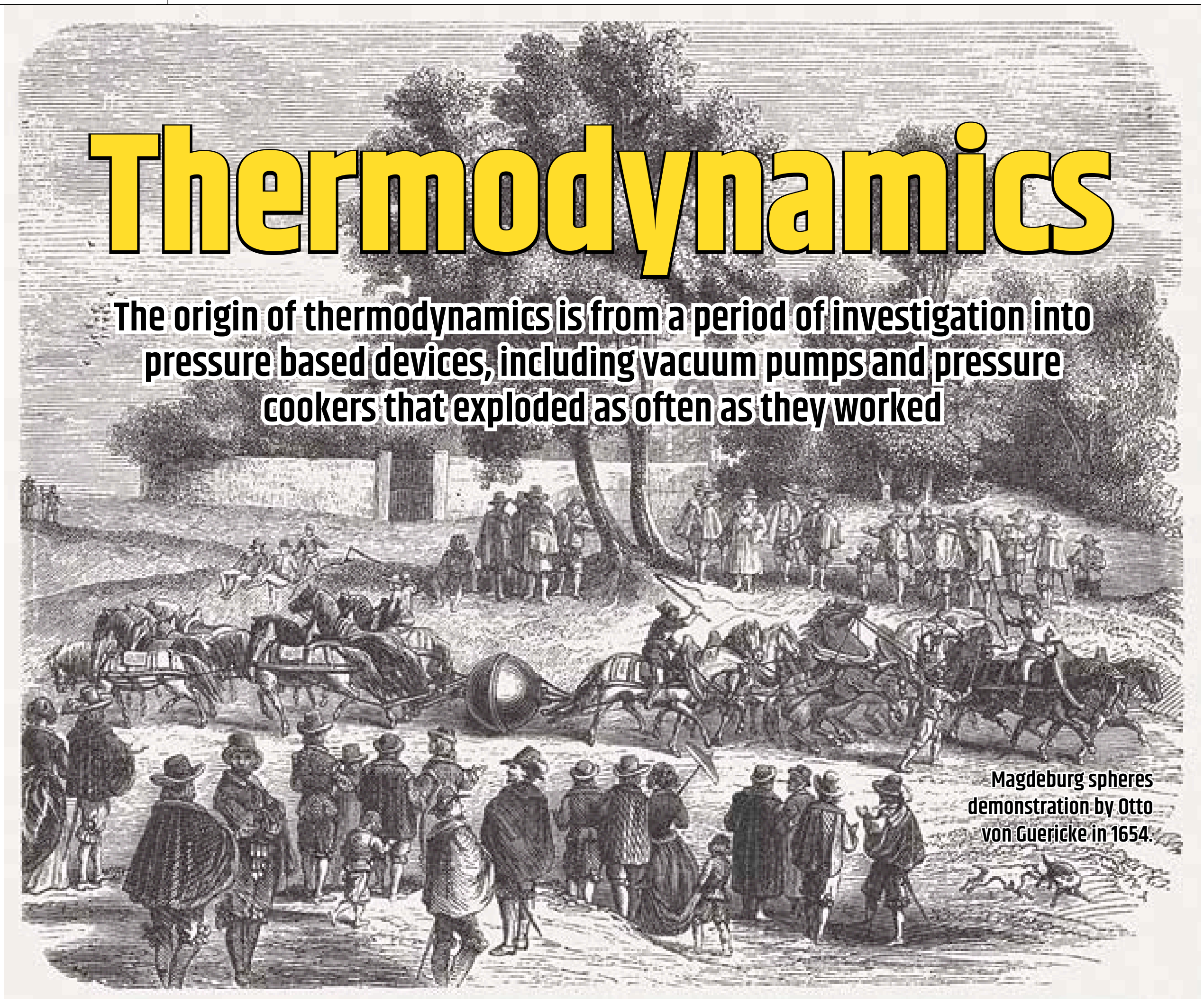
String theory solves mystery
A paper by Kavli IPMU researchers explains how particles behave outside a black hole photon sphere. **http://dgit.in/apr21-55**



Radioactive molecules
Researchers at Caltech propose a new tabletop-based tool to search for answers to the antimatter riddle. **http://dgit.in/apr21-56**

Thermodynamics

The origin of thermodynamics is from a period of investigation into pressure based devices, including vacuum pumps and pressure cookers that exploded as often as they worked



Magdeburg spheres demonstration by Otto von Guericke in 1654.

Aditya Madanapalle | aditya@digit.in

Otto von Guericke was flummoxed by the propositions of Copernican cosmology and could not imagine how bodies of matter could move through vast swathes of free space, where light could travel but sound could not. He embarked on a healthy investigation of creating nothingness, or vacuum, and in this pursuit first tried to burn away all the water from a wooden barrel. He noticed that the air would enter the barrel, and so focused on removing air from containers to creat a vacuum. After a series of experiments, Guericke hacked together an air gun cylinder and a piston into the first functional vacuum pump. There were two way flaps on the pump that allowed air to be pulled out of various containers. Guericke went on to conduct many experiments on containers, and held dramatic public demonstrations to show off the power of air pressure. From 1954, he began to use what are now known as Magdeburg hemispheres, carefully machined copper hemispheres 50 centimeters in diameter, that fit flush against each other. When the air from within them were evacuated, not even two teams of horses could muster enough strength to pull apart the spheres. It is not clear exactly how close to a perfect vacuum the apparatus managed to create. The experiment disproved an ancient proposition by Aristotle, known as horror vacui, or that nature abhors a vacuum. The Magdeburg spheres were also used in demonstrations where they were suspended, with a large number of weights continuously added to dislodge the two halves. Till a valve was turned to equalise the air pressure, the hemispheres would not separate. This remains a dramatic and popular demonstration to this day, with many copies of the magdeburg spheres made on all scales, and the experiment is the inspiration for the Fevicol logo.

The English alchemist Robert Boyle learnt of the work by Guiericke, and with his assistant, Robert Hooke developed a new and improved version of the air pump, known as the machina Boyleana. Hooke was primarily responsible for constructing, operating and demonstrating the apparatus, which was used by Boyle to conduct his experiments. The pump was based on a design by Ralph